SynAthlete-AIoT: An Honest and Explainable AIoT Pipeline for Athlete Injury-Risk Prediction with Synthetic Wearable Data

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Abstract

Wearable artificial-intelligence-of-things (AIoT) systems promise to warn coaches before an athlete is injured, but most published frameworks describe the architecture without building the prediction layer, and reported accuracies are often inflated by evaluation shortcuts such as data leakage. This paper presents SynAthlete-AIoT, a complete pipeline that joins five components: class-conditional generators for synthetic wearable data, simulated edge, fog and cloud processing tiers, a fused sequence-plus-tabular injury-risk model, explanations that are checked by faithfulness tests, and a coach dashboard. Because no public dataset pairs high-frequency body signals with real injuries, the injury model is trained and evaluated only on real daily training logs of 74 competitive runners, while public wearable datasets drive the IoT tiers and the generators. Evaluation follows a protocol written before any experiment: every split is by person, all tuning and thresholds use validation runners only, and ten runners were sealed and opened only once, after the code was frozen. Learned models predict injuries only modestly above chance (ROC-AUC 0.58–0.63 across five development rounds), and the declared fused model kept the same level on the sealed runners. None of the generators produced data that a classifier could not separate from real recordings, and synthetic injury examples gave no measurable gain. Processing at the edge reduced the data sent to the cloud by 593× for the wearable stream, and the explanations passed deletion, randomisation and stability tests in every round. We report every model, including negative results, so that the system can be reused and its limits are clear.

Keywords: Athlete monitoring; Data leakage; Edge computing; Explainable artificial intelligence; Injury prediction; Internet of Things; Synthetic data; Wearable sensors

1 Introduction

Sports injuries rarely appear without warning. They usually follow weeks in which training load, intensity and recovery drift away from what an athlete is used to. A coach who could see this

drift early could adjust the plan before tissue fails. Artificial intelligence has therefore become a popular tool in sport and health science, but recent reviews also warn that many claims rest on small samples and optimistic evaluation [1]. Injury prediction is especially hard: injuries are rare, they depend on many interacting factors, and the relation between load and injury is non-linear and changes over time [2].

Two technical trends make an AIoT approach attractive. First, wearable sensors can record heart rate and motion continuously and with little effort for the athlete [3, 4], and deep networks can recognise activities from such signals [5, 6]. Second, edge and fog computing let devices near the athlete process data locally and send only short summaries to the cloud, which saves bandwidth and energy [7, 8]. Recent designs for athlete health combine these ideas, but they often stop at the architecture and leave the prediction and explanation layers unbuilt or untested.

A practical obstacle is data. To our knowledge, no public dataset pairs high-frequency physiological streams with real injury outcomes. Daily training logs with real injuries do exist [9, 10], while rich wearable datasets [11, 12] contain no injuries. A second obstacle is evaluation. When data from one person appears in both training and testing, or when preprocessing quietly uses test labels, results can look far better than they really are; such leakage has been identified in many fields that use machine learning [13]. With rare events, the choice of metric and the way the data are split also change the conclusion [14–16].

This paper addresses both obstacles. We treat the two kinds of data as two tracks with separate roles, and we evaluate everything under a protocol fixed in advance. The contributions are:

1. An end-to-end, open AIoT pipeline in which every component and every dataset has one explicit, honest role: injury evidence comes only from real injury labels.

2. A person-level evaluation protocol with five development rounds, a sealed final test opened once after a recorded code freeze, a leak alarm, runner-level uncertainty intervals and paired model comparisons.

3. A rebuild of a published injury-prediction method [9] that passed a pre-registered sanity check, together with a label-free variant that a deployed system could use.

4. A comparison of three deep generators and classic augmentation, judged against criteria written before any run, on two wearable datasets and as augmentation of injury data.

5. Explanations that are tested for faithfulness and compared with expectations written before running, instead of being trusted by default.

Section 2 reviews related work, Section 3 describes the proposed methodology, Section 4 reports the results, and Section 5 concludes and outlines future work.

2 Related Work

2.1 Injury prediction and training load

Machine learning has been used to relate screening tests and training records to later injuries. In elite youth football, tree-based models improved the understanding of injury risk but still showed limited predictive accuracy [17]. In elite youth ski racing, training-load characteristics were related to injury and illness risk in a prospective design [18]. Load ratios such as the acute-to-chronic workload ratio are widely used, but alternative load measures have been proposed because the ratio has statistical weaknesses [19]. Two further observations shaped our design: injury risk behaves like a non-linear dynamic system [2], and a previous injury strongly predicts a later one [20]. The second point matters for honest evaluation, because a model that sees a runner's own injury history can appear accurate without learning anything about load. Lövdal et al. predicted injuries of competitive runners from daily logs with bagged gradient-boosted trees [9]; we use their public data [10].

2.2 Wearable sensing and activity recognition

Wearable systems monitor posture, motion and physiology during exercise [3], and machine learning on wearable signals can detect states such as stress in daily life [4]. Deep learning has become the standard approach for sensor-based activity recognition [5], and convolutional networks learn useful features directly from raw inertial signals [6]. We use the PAMAP2 activity dataset [11] for the edge-tier activity model and the WESAD dataset [12] to test whether generators generalise to a second set of people and sensors.

2.3 Edge, fog and cloud computing

Edge and edge–cloud computing move part of the computation close to the sensor, which reduces latency and network load in cyber-physical systems [7]. Fog computing adds an intermediate layer between devices and the cloud and is widely studied for the industrial Internet of Things [8]. Our tiers follow this layering: a reflex check on each window at the edge, session summaries in the fog, and the daily injury model in the cloud.

2.4 Synthetic time-series data

Generative adversarial networks (GANs) learn to produce samples that resemble a training set and have been applied to images, text and signals [21]. For multivariate time series, GAN-based models have been used to impute missing values [22], and simpler augmentation methods that mix or distort real series remain strong baselines [23]. Synthetic data are only useful if they are realistic, help a downstream task and do not copy real records, so we test all three properties.

2.5 Evaluation, calibration and explanation

Leakage and selection outside the test loop are a main cause of irreproducible results [13], and the way data are partitioned changes performance estimates [16]. ROC analysis [14] must be complemented by measures suited to rare positives [15], and predicted risks should be calibrated before they are shown to users [24]. Tree ensembles remain strong for tabular data [25], while gated recurrent networks model sequences [26]. Explainable AI is needed for trust and accountability [27], but explanations must themselves be evaluated: removing the inputs that an explanation marks as important should change the prediction more than removing random inputs [28, 29].

3 Proposed Methodology

3.1 Data and roles

Table 1 lists every dataset and its single role. Injury claims come only from the Lövdal runner logs: 74 runners with a day file (7 days $\times$ 10 features per event) and a week file (3 weeks $\times$ 22 features plus 3 relative changes). PAMAP2 provides the wearable stream (heart rate and three inertial units, resampled to 25 Hz and cut into 5.12-s windows); WESAD tests whether the generators generalise; Sport DB 2.0 checks the realism of synthetic heart rate; two further datasets of unknown origin appear only as labelled demonstrations. The wearable participants are different people from the runners, so the IoT results are systems evidence, not injury evidence. Synthetic data is never treated as evidence about real injuries.

Table 1. Datasets and their single role

Dataset	Origin	Role
Lövdal runners (74 runners, 7 years of daily logs)	real	the only injury evidence: baselines, combined model, explanations, coach view
Kaggle 'Injury Prediction for Competitive Runners'	real (copy)	byte-identical copy of Lövdal: no separate role
PAMAP2 (9 people, wearable sensors)	real	edge sensor stream, activity recognition, generator training and fidelity
WESAD (15 people, wrist sensors)	real	second dataset for generator generalisation
Sport DB 2.0 (55 heart-rate recordings)	real	heart-rate realism check and second edge stream
SoccerMon, PMData	real	exploratory only (Track C), never main evidence
SIRP-600	unknown origin	demo only: shows the SHAP code runs
ziya07 wearable	unknown origin	demo only: dashboard demo stream

3.2 System architecture

Fig. 1 shows the architecture. In Track B, a wearable stream enters the edge tier, which checks every window with a heart-rate and motion rule and an activity network and sends a short summary instead of the raw window. The fog tier groups summaries into sessions, and the cloud tier stores sessions and runs the daily injury model. A cloud-only baseline that sends every raw window is used for comparison. In Track A, the runner logs are split by person; a sequence arm and a tabular arm are combined by a stacking model, calibrated and turned into a 0–100 risk score with an alert level. Generators, explanations and the coach dashboard support both tracks, and the evaluation protocol applies to all of them.

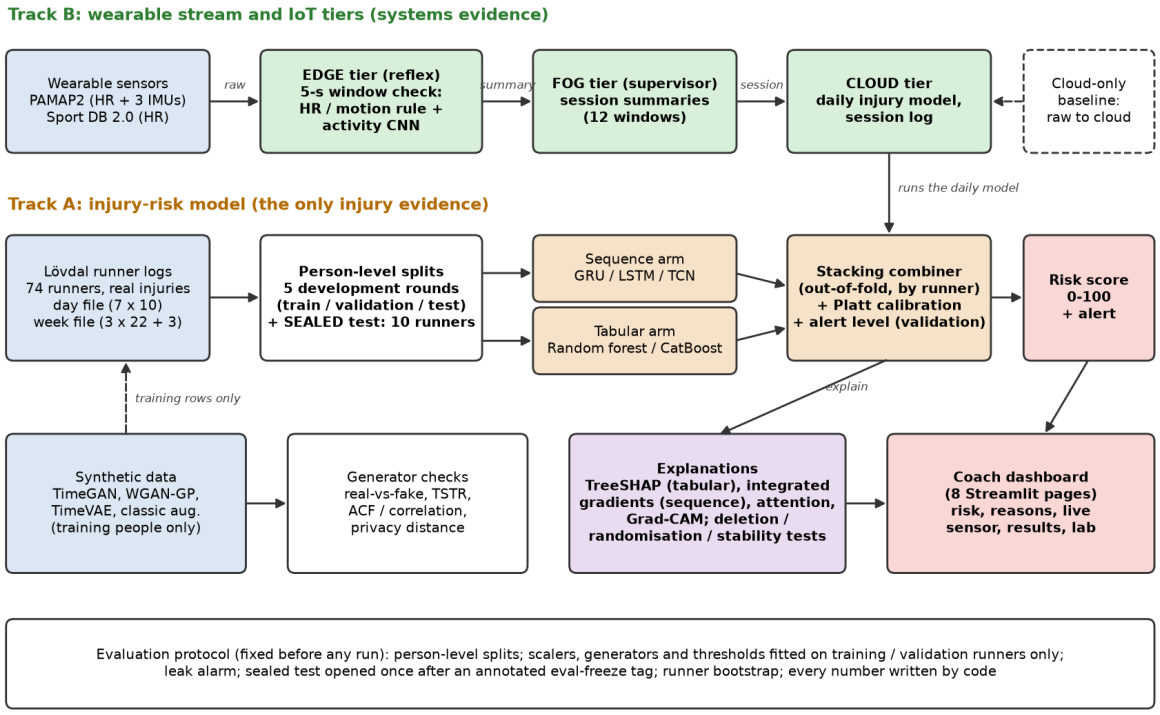


Figure 1. Architecture of SynAthlete-AIoT. Track B carries the wearable stream through the edge, fog and cloud tiers; Track A turns real runner logs into a calibrated 0–100 injury-risk score. Synthetic data enter training rows only, and the sealed test runners are opened once.

3.3 Evaluation protocol

Splits. All splits are by person. The ten most recently joined runners form a sealed final test. The other 64 runners form five development rounds; in each round, settings, early stopping, calibration and the alert level use that round's validation runners only, and its test runners are scored once. The sealed runners can be loaded only through a guarded function that checks an annotated, pushed `eval-freeze` tag and an unchanged code tree, and it records every opening.

Leakage controls. One z-score scaler is fitted on training runners only. Identity, date and own-history inputs are excluded, because a runner's own event history reveals the label in this

dataset. A shuffled-label model acts as a leak alarm; its ROC-AUC stayed at 0.498 (day) and 0.513 (week), i.e. at chance. Independent code reviews were run before the freeze and before reporting.

Metrics. For a test set with injured events P and healthy events N, ROC-AUC is the probability that a random injured event receives a higher score than a random healthy one,

$$\mathrm{AUC} = \frac{1}{|P|\,|N|} \sum_{i \in P} \sum_{j \in N} \mathbb{1}[s_i > s_j], \tag{1}$$

and PR-AUC is the average precision over recall levels, whose chance level equals the injury rate. PR-AUC is the headline metric because injuries are rare [15]. Calibrated probabilities are obtained by Platt scaling fitted on validation runners,

$$\hat{p} = \frac{1}{1 + \exp\big(-(a \operatorname{logit}(s) + b)\big)}, \tag{2}$$

and the risk score shown to the coach is the percentile of $\hat{p}$ among the round's validation events. The alert level minimises |sensitivity − specificity| on validation runners.

Uncertainty and comparisons. Intervals come from a runner-level bootstrap with 2,000 resamples, resampling whole runners within each test group. Models are compared in predeclared pairs on the same rows; a difference counts only when its 95% interval excludes zero. The intervals are not adjusted for multiple comparisons, which we state wherever they are used.

3.4 Injury models

The baselines are constant risk, an acute-to-chronic style rule, logistic regression, random forest, CatBoost and a gated recurrent unit (GRU) network. The Lövdal method was rebuilt twice: as an exact copy, including its normalisation step that reads the labels, which was run once as a sanity check on stand-in runners; and as a label-free variant compared on all five rounds. Each tunable model tries eight settings per round, drawn by a seeded random search from grids written before the first run; the chosen setting is refitted with three seeds.

The **fused model** combines a sequence arm (GRU, LSTM or temporal convolutional network) on day sequences with a tabular arm (random forest or CatBoost) on week features. Both arms are chosen on validation runners. Their predictions are combined by a logistic-regression stacker trained on out-of-fold predictions grouped by runner, so the combiner never sees a prediction made by a model that trained on the same runner. With 44 tries per round, the fused model has the same tuning budget per component as every baseline, and it was declared in advance as the model for the sealed test.

3.5 Synthetic data generators

Class-conditional TimeGAN, WGAN-GP and TimeVAE are trained only on each round's training people; settings are chosen by validation train-on-synthetic, test-on-real (TSTR) macro-F1, except WGAN-GP, which uses one declared setting and is reported as untuned. Classic augmentation (jitter, scaling and time-warping) is the baseline. Four criteria were fixed in advance: a real-versus-synthetic classifier should be close to 50% accuracy (within 0.15), TSTR macro-F1 should be within 0.15 of real-data training, the autocorrelation and cross-channel correlation errors should be below 0.10, and at most 5% of synthetic windows may lie closer to a real training window than real test windows usually do (a privacy check). For injury augmentation, a conditional TimeGAN adds synthetic injury sequences to training rows only; validation and test data stay real.

3.6 Edge, fog and cloud tiers

The three tiers run as separate processes connected by queues. All processes finish their set-up before the first window, and windows are released at a steady pace of one every 0.1 s, so that the measured latency is decision time rather than start-up waiting. We record the bytes that reach the cloud and the median decision latency, for a PAMAP2 test person and a Sport DB 2.0 heart-rate recording, in tiered and cloud-only mode, with three repetitions each.

3.7 Explanations

TreeSHAP explains the tabular arm and integrated gradients explain the sequence arm. An explanation-only attention model is always shown next to integrated gradients, never alone, and one-dimensional Grad-CAM explains the activity network. Three tests with thresholds fixed in advance check faithfulness: deletion (removing the top inputs must lower the risk at least 1.5 times more than removing random inputs), model randomisation (explanations of a model with random weights must differ, rank correlation below 0.5) and stability (near-identical inputs must give near-identical explanations, rank correlation above 0.8). Before running, we also wrote down which inputs we expected to raise risk: recent load spikes, monotony and low recovery. Each risk score is turned into a plain-language reason for the coach.

4 Results

4.1 Injury prediction

Table 2 and Fig. 2 report every injury model on runners never seen in training. The leak alarm stayed at chance. Learned models reach ROC-AUC 0.58–0.63, clearly above chance but far from reliable prediction. The fused model reaches ROC-AUC 0.609 ± 0.024 and PR-AUC 0.022 ±

Table 2. Injury prediction, mean ± SD over five development rounds (chance PR-AUC ≈ injury rate)

Model	File	PR-AUC	ROC-AUC
Constant risk	day	0.013 ± 0.000	0.500 ± 0.000
Constant risk	week	0.013 ± 0.000	0.500 ± 0.000
ACWR-style rule	week	0.015 ± 0.001	0.540 ± 0.021
Logistic regression	day	0.022 ± 0.004	0.626 ± 0.019
Logistic regression	week	0.020 ± 0.004	0.621 ± 0.029
Random forest	day	0.019 ± 0.002	0.603 ± 0.024
Random forest	week	0.022 ± 0.007	0.614 ± 0.042
CatBoost	day	0.021 ± 0.004	0.598 ± 0.043
CatBoost	week	0.019 ± 0.002	0.606 ± 0.028
GRU	day	0.023 ± 0.006	0.631 ± 0.009
Lövdal, label-free	day	0.017 ± 0.003	0.584 ± 0.044
Lövdal, label-free	week	0.021 ± 0.006	0.589 ± 0.066
Fused	shared	0.022 ± 0.003	0.609 ± 0.024

0.003. The exact copy of the published method passed its pre-registered sanity check (day-file ROC-AUC 0.617 (95% CI 0.573–0.664)), and the label-free variant reaches ROC-AUC 0.584 (day) and 0.589 (week). Because the two were scored on different runners, the effect of the label-reading step was not measured head-to-head.

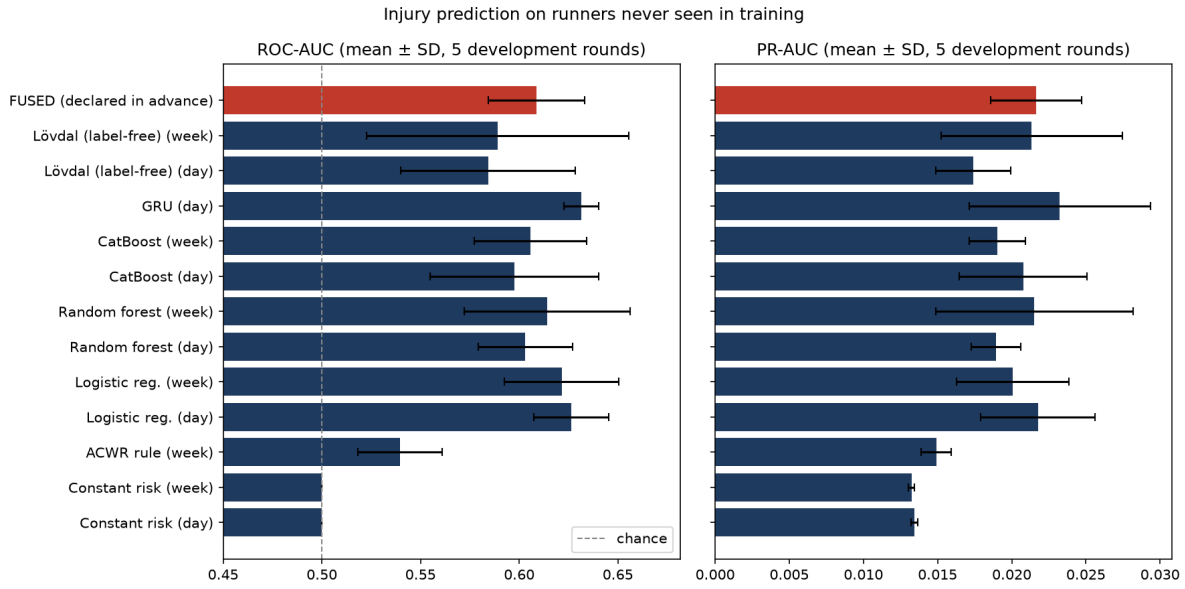


Figure 2. ROC-AUC (left) and PR-AUC (right) of every injury model, mean and standard deviation over five development rounds. The fused model is highlighted because it was declared in advance, not because it was best.

Fig. 3 shows what an alert means in practice. At the alert level chosen on validation runners, the fused model catches a majority of injuries, but it also raises many false alarms, because injured and healthy days receive overlapping scores. As an early-warning tool, it is therefore a prompt for a conversation with the athlete rather than a diagnosis.

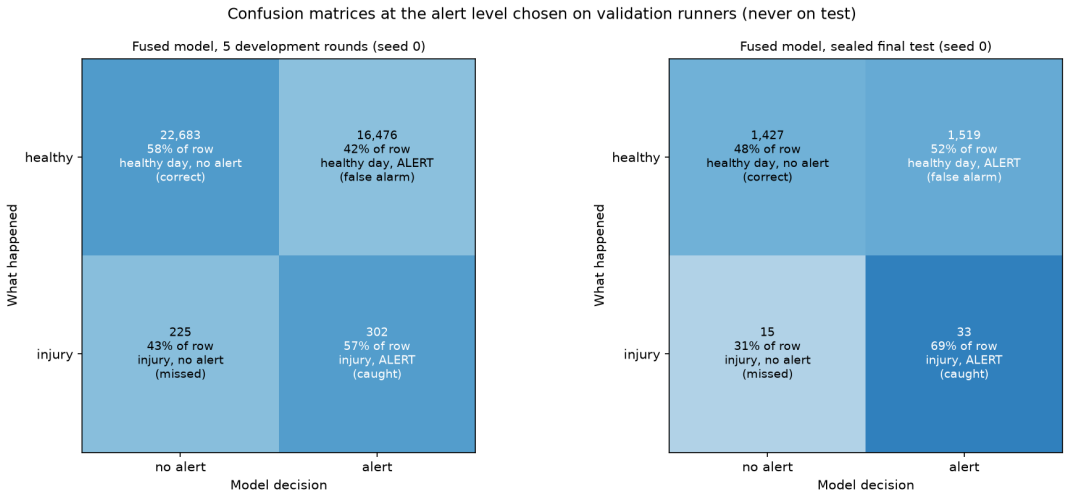


Figure 3. Confusion matrices of the fused model at the alert level chosen on validation runners: pooled test runners of the five development rounds (left) and the sealed final test (right).

4.2 Sealed final test

On the 10 sealed runners (2,994 shared days, 48 injuries in 7 runners), opened once after the evaluation freeze, the declared fused model reached ROC-AUC 0.605 (0.535-0.699) and PR-AUC 0.030 (0.016-0.056; chance 0.016); its alert level caught 65% of injuries with 46 false alarms per caught injury. Secondary results, scored once and not a comparison (other rows and more training runners): the label-free variant reached ROC-AUC 0.649 (day) and 0.602 (week); no model was switched. With so few runners the intervals are wide and understate the uncertainty. Fig. 4 places this result next to the development estimate. The two agree, which indicates that the development results were not inflated by tuning. The secondary results are shown for completeness only; they were scored on different rows and with more training runners, so they are not a ranking of models.

4.3 Model comparisons

Table 3 and Fig. 5 give paired differences with runner-bootstrap intervals. Against the label-free Lövdal variant, logistic regression and the GRU on the day file are clearly better on both metrics, and CatBoost on the day file on PR-AUC only; constant risk and the acute-to-chronic style rule are clearly worse. The fused model's gain over its sequence arm is borderline on ROC-AUC and absent on PR-AUC, and it shows no gain over its tabular arm. With 24 unadjusted intervals, borderline results should be read with care.

4.4 Synthetic data

Table 4 and Fig. 6 compare the generators with real-data training (real-only macro-F1 0.632). A real-versus-synthetic classifier separated the synthetic clips of every generator easily, so no

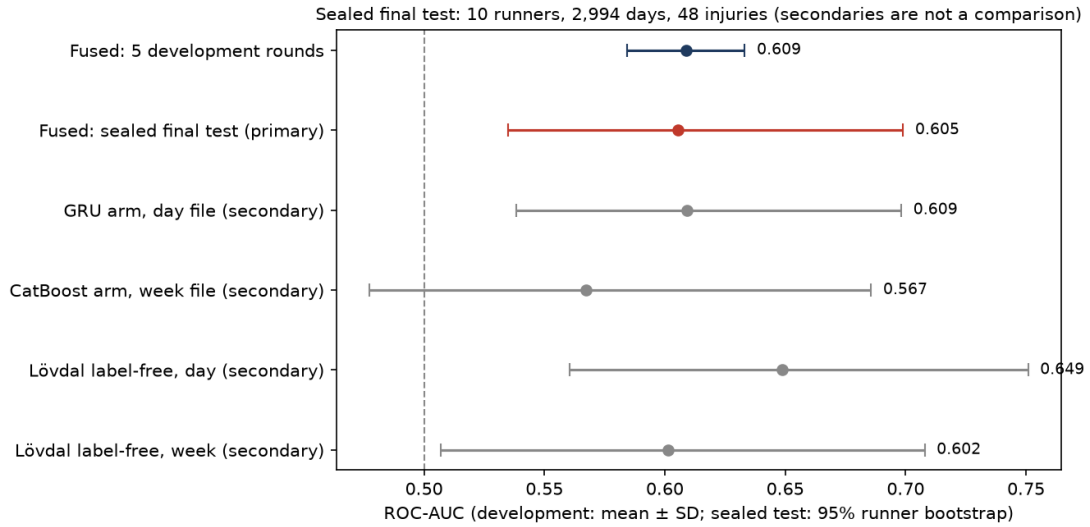


Figure 4. Sealed final test. Development: mean ± standard deviation over five rounds; sealed test: 95% runner-bootstrap interval. Grey rows are secondary results and are not a comparison.

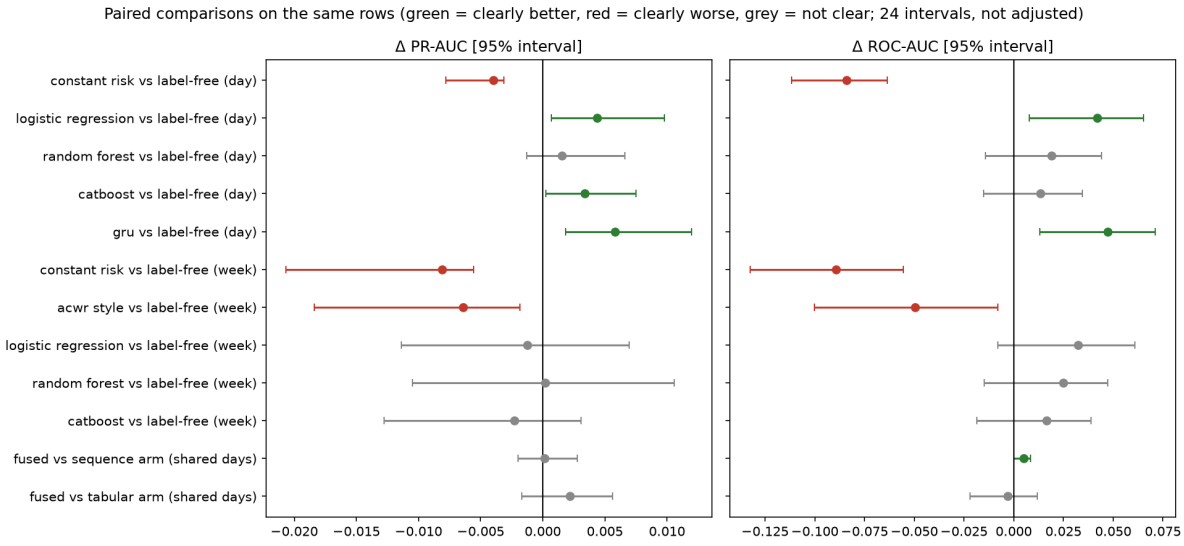


Figure 5. Paired differences on the same rows with 95% runner-bootstrap intervals (green: clearly better; red: clearly worse; grey: not clear).

Table 3. Paired differences (5-round mean, runner bootstrap 95% interval, 2,000 resamples)

Comparison	Δ PR-AUC	Δ ROC-AUC
constant risk vs label-free (day)	-0.0040 [-0.0078, -0.0031]	-0.084 [-0.112, -0.064]
logistic regression vs label-free (day)	+0.0043 [+0.0007, +0.0098]	+0.042 [+0.008, +0.065]
random forest vs label-free (day)	+0.0015 [-0.0013, +0.0066]	+0.019 [-0.014, +0.044]
catboost vs label-free (day)	+0.0034 [+0.0002, +0.0075]	+0.013 [-0.015, +0.034]
gru vs label-free (day)	+0.0058 [+0.0018, +0.0120]	+0.047 [+0.013, +0.071]
constant risk vs label-free (week)	-0.0081 [-0.0206, -0.0056]	-0.089 [-0.132, -0.056]
acwr style vs label-free (week)	-0.0064 [-0.0184, -0.0019]	-0.049 [-0.100, -0.008]
logistic regression vs label-free (week)	-0.0013 [-0.0114, +0.0070]	+0.032 [-0.008, +0.061]
random forest vs label-free (week)	+0.0002 [-0.0105, +0.0106]	+0.025 [-0.015, +0.047]
catboost vs label-free (week)	-0.0023 [-0.0128, +0.0031]	+0.017 [-0.019, +0.039]
fused vs sequence arm (shared days)	+0.0001 [-0.0020, +0.0028]	+0.005 [+0.000, +0.009]
fused vs tabular arm (shared days)	+0.0022 [-0.0017, +0.0056]	-0.003 [-0.022, +0.012]

Table 4. Generators on PAMAP2 (4 rounds): discriminative accuracy (0.5 = indistinguishable), TSTR and real+synthetic macro-F1, pre-registered criteria passed

Generator	Disc. acc.	TSTR	Real+syn.	Passed
timegan	0.956	0.270	0.639	0/5
wgan_gp (untuned)	0.898	0.564	0.637	1/5
timevae	0.936	0.525	0.647	2/5
classic	0.860	0.624	0.636	2/5

generator met the realism criterion; training on synthetic data alone came closest to real training for classic augmentation, and adding synthetic clips to real ones changed little. On WESAD (five folds, settings carried over from PAMAP2 without tuning), TimeGAN met only the privacy criterion of its 4 fidelity and privacy criteria; WGAN-GP met none of its 4 fidelity and privacy criteria; TimeVAE met none of its 4 fidelity and privacy criteria. Adding synthetic injury examples to the training rows only gave the GRU PR-AUC 0.022 ± 0.003 with TimeGAN and 0.021 ± 0.005 with classic augmentation, against 0.023 ± 0.006 real-only (ROC-AUC 0.633 ± 0.006, 0.617 ± 0.015 and 0.631 ± 0.009); there is no paired interval, so no improvement is claimed.

4.5 Edge, fog and cloud tiers

Table 5 and Fig. 7 show the tier benchmark. Processing at the edge reduced the data sent to the cloud by 593× for the wearable stream and by 85× for the heart-rate stream. Decisions took a few milliseconds in both modes; because everything ran on one computer, there was no network delay to save, so latency compares processing only and the data saving is the main result.

4.6 Explanations

Out of five rounds, the tabular-arm SHAP explanations passed the deletion test in 5, randomisation in 5 and stability in 5; the sequence-arm integrated gradients passed 5, 5 and 5. Against the expectations written before running, “recent load raises risk” agreed in 10 of 10 arm-rounds;

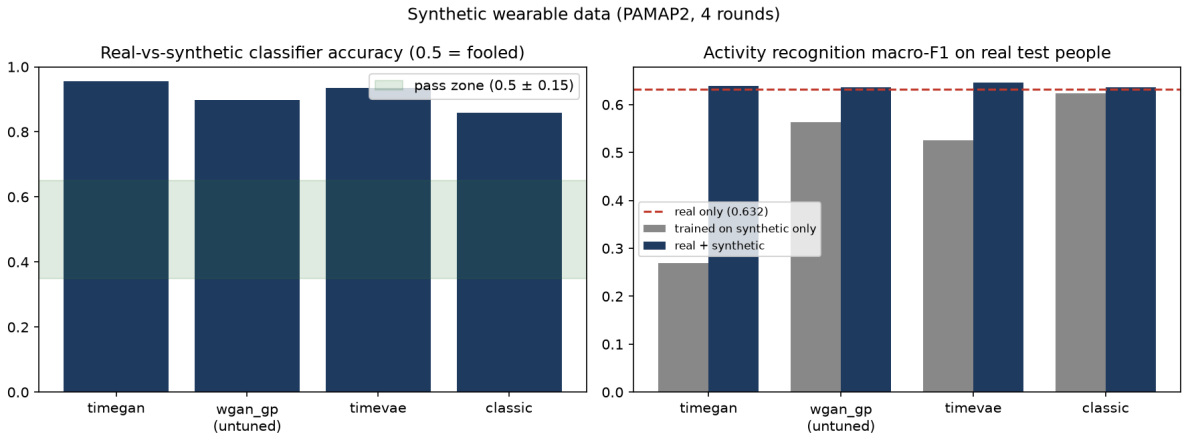


Figure 6. Generators on PAMAP2 (four rounds): real-versus-synthetic classifier accuracy with the pass zone (left), and activity-recognition macro-F1 when training on synthetic data only or on real plus synthetic data, against real-only training (right).

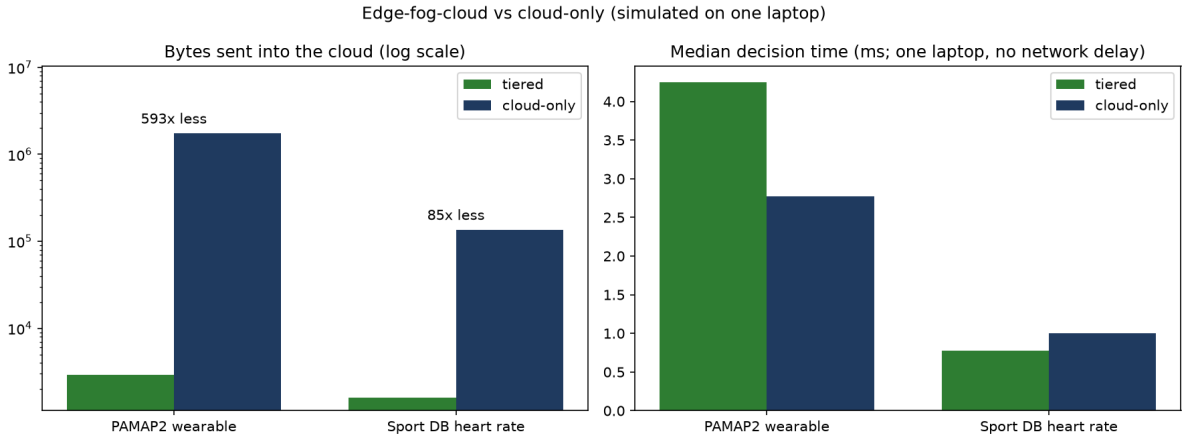


Figure 7. Bytes sent into the cloud (left, log scale) and median decision time (right) for the tiered design and a cloud-only design, simulated on one computer.

Table 5. Edge–fog–cloud vs cloud-only (simulated on one computer)

Stream	Mode	Bytes into cloud	Median decision (ms)
pamap2	tiered	2,932	4.25
pamap2	cloud_only	1,739,186	2.77
sportdb2_hr	tiered	1,600	0.78
sportdb2_hr	cloud_only	135,619	1.00

"low recovery raises risk" agreed in 2 of 10 arm-rounds (Fig. 8). Attention weights and integrated gradients often disagreed, which supports showing attention only next to a tested method.

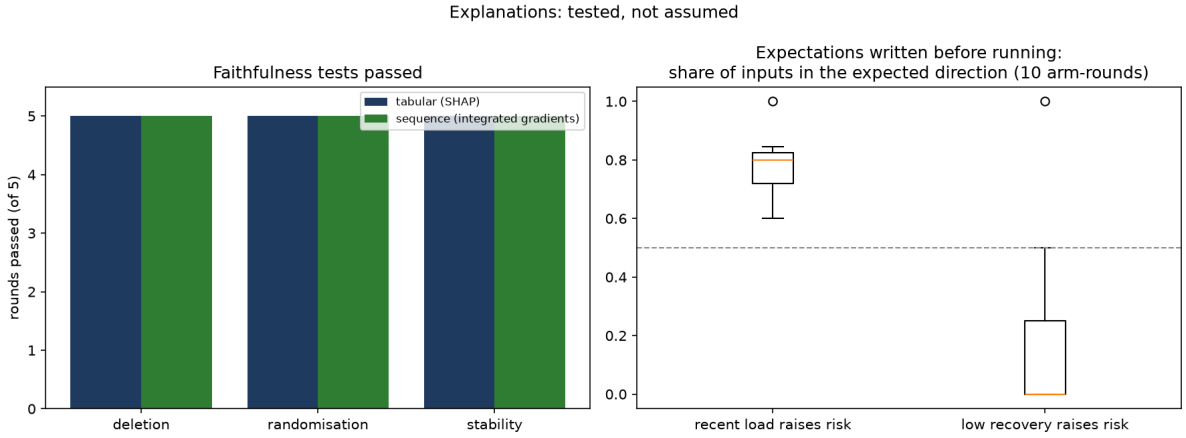


Figure 8. Faithfulness tests passed per explanation method (left) and agreement with expectations written before running (right).

4.7 Discussion

Three findings stand out. First, daily training logs carry only a weak signal before an injury: all honest models sit modestly above chance, and the alert level that balances sensitivity and specificity raises many false alarms. Second, the evaluation design matters: the sealed test reproduced the development estimate, and paired intervals showed that most apparent differences between models are not reliable. Third, the IoT design delivers a clear, measurable benefit in data volume even though the injury signal is weak. The main limitations are the small sealed test (ten runners), the removal of healthy days close to injuries by the dataset's authors (so alert rates are not daily rates), the simulation of the tiers on one computer, and the fact that the wearable participants are different people from the runners.

5 Conclusion and Future Scope

We built and honestly evaluated a complete explainable AIoT pipeline for athlete injury risk. Under a person-level protocol with a sealed final test, injury risk from daily training logs is

predictable only modestly, the result held on runners nobody had looked at, the edge tier sharply reduced cloud traffic, and the explanations passed faithfulness tests. Synthetic wearable data did not reach realism and did not improve injury prediction, which we report as a negative result rather than hide.

Future work has five directions. (i) Collect a dataset that pairs wearable signals with real injuries from the same athletes, so that physiology can be used as injury evidence. (ii) Deploy the tiers on real devices and networks to measure energy use and network latency. (iii) Validate the model prospectively on new athletes and new seasons, with larger cohorts. (iv) Improve the generators, for example with privacy-aware training and evaluation on larger datasets. (v) Study with coaches how risk scores and reason sentences change training decisions.

Acknowledgement

We thank our guide and the Department for their support. AI coding assistants helped write code and text under the authors' direction; every AI contribution is logged in the project's AI log, every number is produced by code from saved result files, and all claims were checked by the authors.

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